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Research

A Deadly Cure: Animal Venoms and Toxins in Drug Development

Life stems from death. The decomposition of one organism provides the resources and conditions for new life to flourish, a notion central to ecosystems and the universal laws of mass and energy. Thereupon, what if we were to indulge in this paradox and employ it to our advantage? That is the question scientists and researchers have been asking themselves for decades now, all while investigating what I like to call a *deadly* cure. Specifically, they have been exploring the violent nature of flesh-destroying compounds such as animal venoms for the purpose of creating life-saving drugs. In this essay, I present a research-based overview of the applications of venom and toxin-derived drugs in medicine, as well as the productive and ethical challenges they face. Which, furthermore, offers an optimistic and proliferative future in the healthcare world.

Venom is a potent biochemical weapon evolved by numerous species for predation, defense and competition; it has long been a subject of human fear and fascination. Comprising a complex cocktail of proteins, peptides, and other molecules, venoms exhibit highly specific interactions with physiological targets, leading to dreadful toxic effects in the affected organism. Yet, snake venoms have been used in traditional medicine for thousands of years. According to Dr. Liji Thomas, they used to be the basis of preparations for treating smallpox, leprosy, and healing wounds for civilizations ranging from the Greeks to the 19th

century with Albert Calmette's revolutionary work¹. Since, venoms have been used for the development of treatments for a myriad of medical conditions, transforming agents of harm into instruments of healing.

Venom-derived compounds have revolutionized modern medicine by offering innovative solutions to health problems. One remarkable example is the Gila monster. Apart from the lizard's beautifully spotted orange and black armor, we can find among their jaws and saliva a hormone called exendin-4. Marvelously, evolution has given these dragon-like reptiles the ability to withstand living in deserts and arid scrublands by developing this compound (Ying, 2013). The hormone slows down digestion and food absorption, keeping Gila monsters alive in the race even when their hunting brings them no food for a prolonged amount of time. Moreover, for the benefit of 830 million people, exendin-4 also increases insulin production. Scientists have been able to mimic it in the synthetic form of exenatide, which imitates the action of glucagon-like peptide-1 (GLP-1). This compound enhances glucose-dependent insulin secretion, helping individuals with type 2 diabetes improve their glycemic control and lose weight, as shown in clinical trials (Triplitt, 2003). Recent research even suggests that exendin-4 may be a potential treatment for neurodegenerative diseases such as Alzheimer's or Parkinson's (Verma, 2023) among other miraculous uses including the prevention of ischemic strokes (Teramoto, 2015). The drug's development underscores the untapped medical potential in natural toxins and venoms, as it transforms harmful substances into life-saving treatments.

As promising as the Gila Monster lizard may seem, most US Food and Drug Administration (FDA) and European Medicines Agency (EMA) approved drugs are derived

¹ Albert Calmette (1863-1933) was a French physician, bacteriologist, and immunologist. He developed the first antivenom serum for snakebites in 1894; his research laid the foundation for modern antivenom therapy. Calmette is also recognized for co-developing the BCG vaccine for tuberculosis. He was nominated for the Nobel Prize in Medicine thrice (1928, 1930, and 1932).

from snake venoms. This is due to the larger number of venoms produced by snakes in comparison to smaller animals such as scorpions, spiders, and snails. In fact, the effect of snake venoms on hemostasis² has presented us with the cardiovascular system as a major pharmacological target for snake venom toxins. Moreover, the analytical techniques capable of characterizing limited amounts of venom components from small animals were only developed recently (Bordon, 2020). These challenges initially drove toxicology research towards a focus on snake venoms. Even though recent advancements in technology and improved experimental models have enabled the investigation of low-abundance toxins and pharmacological targets, snake venoms remain the predominant source of investigation.

The journey of snake venom-derived drugs began with captopril, an angiotensin-converting enzyme (ACE) inhibitor used to treat hypertension and certain cardiovascular conditions. Developed in the 1970s by Squibb³, captopril was inspired by a peptide found in the venom of the Brazilian pit viper (*Bothrops jararaca*). Renowned in the wealthy and densely populated areas of south-eastern Brazil, the snake's bite is highly venomous and leads to horrific swelling, systemic bleeding, organ failure, and even death if left untreated with antivenin. Millions of years of evolution have gifted this viper's venom with peptide bradykinin-potentiating peptides (BPPs); complex as their name, they function by causing the threat or prey to lose consciousness by briskly dropping their blood pressure. In a healthy person, the consumption of this compound would result in the gruesome effects listed previously; but for someone suffering from elevated blood pressure, the outcome would seem to be rather helpful. This peptide inhibits ACE, an enzyme responsible for converting angiotensin I to the vasoconstrictor angiotensin II, basically regulating fluid balance in the

² Hemostasis is the body's physiological process of stopping bleeding through the formation of blood clots.

³ John Vane joined by Sergio Henrique Ferreria and his team were responsible for the discovery of captopril. In the 1980's, Vane was awarded the Nobel Prize in Medicine and was knighted for his work. Ferreira received the National Order of Scientific Merit from Brazil.

body by expanding blood vessels and lowering blood pressure (Marte, 2022). After recreating BPPs in the form of captopril, cardiovascular treatment for patients prone to heart failure, hypertension, post-MI left ventricular dysfunction, and diabetic nephropathy was revolutionized (Thomas, 2021). Nonetheless, as a drug, captopril was still far from perfect. To resolve the side effects of captopril caused by its thiol group, Merck shaped a new and improved version.

Captopril's success led to the development of enalapril, a related ACE inhibitor with improved pharmacokinetic⁴ properties. The thiol group causing a metallic taste, dizziness, and cough in captopril was substituted by carboxylate. Even though this resulted in a loss of potency, it was compensated for with additional modifications such as the removal of a sulfhydryl group. The product, enalaprilat, continued to undergo improvement until its conversion gave birth to enalapril; it resulted in having very good oral bioavailability⁵ and was approved by the FDA and EMA. Consequently, enalapril established itself as Merck's first billion-dollar-selling drug in 1988. (Oliveira, 2022). As time has passed, multiple other ACE-inhibitor drugs based on enalapril have been developed and approved. Some examples include lisinopril, quinapril, and ramipril among others. These medicines have proved themselves as global superstars in markets around the world and showcase how a mere push can trigger a cascade of advancements in immense therapeutic potential (Oliveira, 2022).

Almost ten years after enalapril's feat, a new generation of snake-derived drugs emerged, this time their focus was aimed towards preventing our platelets from clumping together; therefore, receiving the name antiplatelet drugs. Their inspiration was once again derived from the Middle East's hauntingly impressive *Echis carinatus*. With only the

⁴ Pharmacokinetics is the study of how the body interacts with an administered substance during the entire duration of its effects.

⁵ Bioavailability refers to the extent and rate at which an active drug enters the system circulation and accesses the site of action.

insignificant dose of 5mg of venom, the saw-scaled viper can produce a significant mortal outcome in us humans. Despite its deceiving small size along with their sublime saw-like sizzling sound, this viper is responsible for the most amount of snakebites in the world (Rafferty, 2019). Moreover, much like their prolonged riveting scale pattern, their toxin echistatin is composed of 49-residue disintegrins⁶. Echistatin competes with fibrinogen for binding to a specialized integrin (α IIb β 3), which inhibits the final step in platelet aggregation (Olivera, 2022). In a simplified manner, echistatin's structure of amino acids (RGD motif) doesn't allow fibrinogen to join the vascular playground, as it is an essential protein in blood clot formation (Chen, 1991). Tirofiban was modelled by Merck⁷ to replicate this amino acid chain and enhance the compound's binding affinity and specificity. Tirofiban has shown effectiveness and success in reducing ischemic complication in patients with acute coronary syndrome and percutaneous coronary intervention. Therefore, consolidating itself as a major product by combating the most common cause of death in the world, ischemic heart disease, and establishing itself as an example of venom turned into a life-saving drug (WHO, 2024).

In today's ever-shifting world, snake venom is also emerging as a promising tool in cancer research, with its unique bioactive⁸ compounds offering new possibilities for targeted therapies. Certain proteins and peptides such as crotamine derived from venom exhibit cytotoxic properties that can selectively target cancer cells while sparing healthy tissues, a critical advantage in reducing the side effects of conventional treatments (Li et al. 2018). This accuracy is the holy grail of oncology. For example, studies led by Dr. Li demonstrate how venom components from cobras and vipers show potential in treating aggressive cancers,

⁶ Disintegrins are small, non-enzymatic, cysteine-rich proteins, primarily found in snake venom. Disintegrins typically contain specific motifs, such as RGD (Arg-Gly-Asp) or KGD (Lys-Gly-Asp), which allow them to bind to integrins and block their interaction with substances.

⁷ Specifically, by the team comprised of George Hartman, Melissa Egbertson, and Wasyl Halczenko.

⁸ Bioactivity refers to the capacity of a substance or material to elicit or modulate a biological response in an organism.

including breast and colon cancer, by disrupting cancer cell membranes and inhibiting tumor growth (Li et al. 2018). Similarly, a study made by the University of Southern Colorado highlights how researchers are leveraging venom compounds to develop precision therapies that improve patient survival outcomes. Despite these breakthroughs, challenges remain in isolating, refining, and scaling these venom-based treatments for widespread use. It's like panning for gold: we must find a way to protect healthy cells while eliminating cancerous ones. Nevertheless, ongoing advancements in biotechnology and genetics hold promise for revolutionizing and developing novel cancer therapy.

I could easily keep adding onto the list about the marvel potential of these venom derived compounds; however, it is also vital to recognize that venom research is fraught with significant challenges. Finding and utilizing the medical potential engraved within this chemical puzzle can be compared to the grueling extraction process of an arcane diamond buried within Sierra Leone. The first roadblock in its way lays within the ethical and logistical challenges concerning the welfare of captive snakes and other animals. In captivity, ethical concerns arise from the long-term confinement of these creatures. This is because the average lifespan of a snake can extend up to twenty years; even so, their housing conditions often do not align with their natural behaviors which leads to stress and potential issues with their well-being (Halm, 2024). Moreover, many venom labs keep snakes in restrictive enclosures that limit social interactions between mothers and their young, as well as other creatures, raising questions about the balance between scientific necessity and animal rights. Additionally, logistical hurdles include ensuring the maintenance of secure facilities, and training personnel to extract venom from these animals without harming them or themselves. Overall, proper venom collection demands strict biosafety protocols to prevent contamination and ensure the quality of the venom used in research and pharmaceutical applications (Santos et al., 2020). Thus, companies should take measures such as organizations like the Center for

the Study of Venoms and Venomous Animals (CEVAP) who have implemented rigid and standardized procedures regarding the handling of these animals from capture to venom collection (Santos, 2020). Some of these norms include customized social vivariums, appropriate staff training in herpetology and toxicology, and a greater number of captive snakes (decreasing the amount of times venom is extracted each month), and therefore increasing animals' life quality. In the hope of removing moral concerns, investment in improved venom production will not only better workers' and animals' lives alike, but also the future lives of thousands of people.

Still, the sharpest edge of this diamond lies within the synthetic replication of the components that integrate venom and their transformation into medicine. Snakes produce larger venom quantities compared to smaller venomous species, facilitating the study of the function, structure, and interaction of proteins and toxin identification. However, venom's chemical diversity is lower than that of traditional chemical compound libraries used in pharmaceutical research. So, the real advantage of animal venoms and toxins lies within their high specific and inherent bioactivity, which enables a drug-candidate to be chosen through a rational process based on previously observed bioactivities. The cornucopia of diseases we can treat seems to be never-ending; regardless of this, the pharmaceutical industry currently lacks the flexibility needed and inclines toward automation and not rationality (Oliveria, 2022).

There exist two major approaches for developing snake venom-derived drugs: direct use of toxins or the creation of small synthetic analogs called toxinomimetics. The former faces issues related to stability, administration, and large-scale production, leading to limited success in drug approval by the FDA and the EMA. Altogether, the use of unmodified snake toxins has not flourished exactly as planned because it hinders the medicine's administration. On the other hand, toxinomimicry has prevailed and found ways around these issues. It is a

technique that involves complex and rational transformations to the toxin core and has been instrumental in the development of drugs like captopril and eptifibatide. Biomimicry offers improved safety, efficacy, and bioactivity, providing us with a preferable path. Despite this novel method, we must not forget the complexity of venom and the tangled serpentine trail a drug follows as it enters our body (Oliveria, 2020).

After overcoming early research challenges, such as obtaining sufficient quantities of the toxin, further hurdles arise in preclinical and clinical evaluation. Many venom-based compounds struggle to cross biological barriers in our bodies before reaching their target, such as the blood-brain barrier or our immune system (Bordon, 2020). This leads to the rapid degradation of the drug in the body, reducing their efficacy. Besides, preclinical studies are time-consuming and expensive, as they must comply with stringent regulatory requirements worldwide. In many cases, a lack of funding and manufacturing capabilities hinder academic researchers from advancing their drug candidates to the clinical trial phase (Ibraheem et al., 2014). Furthermore, randomized clinical trials are the gold standard for assessing the effectiveness and safety of new drugs before they hit the market shelves. However, these tests have limitations, particularly in evaluating special populations such as children, pregnant women, and the elderly. Factoring this in, achieving regulatory approval is even more challenging since the drug's bioavailability must also compete with current market bioavailability and pharmacokinetics of other products even if they offer a better and more sustainable solution (Bordon, 2020). Adding up to these extremely high standards are off-target effects. If the toxin binds to multiple receptors or interacts with unintended cells, it can lead to unexpected side effects and toxicity within the organism (Harvey, 2014). These adverse reactions, along with insufficient research, are the leading causes of clinical trial failures. Scientists must be atomically precise in the purification of these compounds.

The course of action we must follow to overcome these barriers involves a greater use and understanding of computational chemistry. Leveraging increased computational power, along with deep learning and artificial intelligence, has facilitated significant progress in several key areas. One of the most important applications is protein homology modeling, which allows researchers to predict the structures of venom toxins and their biological targets. A complement to this technology is protein-protein docking, that allows us to forecast how toxins will interact with these targets (Ojeda, 2017). This is comparable to a child learning from feedback every time their parent scolds them; a simulation based on machine learning helps us to refine compounds little by little, until we obtain a mature, adult version of them. Other key tool techniques such as computational mutagenesis, particularly alanine scanning, help us systematically alter amino acids within venom proteins to determine which residues are beneficial for proper function. A notable example of this method is clarifying how mambalgins (venom peptides from black mamba snakes) inhibit sensing ion channels, which are potential drug targets for pain management (Ojeda, 2017). Screening databases of biological targets and bringing into play computer simulations will play a crucial role in identifying the structures and uses of venom in drug development with never-before-seen speed and safety.

In Western culture, serpents are viewed as symbols of deceit and power. Their nature has triggered thoughts of wickedness and fearful fascination among us. Nonetheless, ancient civilizations respected these animals due to their believed healing powers. With modern technological advancements, it has become undeniable that our predecessors were right, the venom of these captivating animals is a boundless library of bioactive compounds with great medicinal potential. The paradox of using death to cure life is elegantly embodied in the study of these compounds. Yet, the path to harnessing them is riddled with ethical, logistical, and scientific challenges. The balance between scientific progress and animal welfare

remains a crucial concern, while venom extraction, drug development, and regulatory approval require immense precision and perseverance. However, advancements in biomimicry, synthetic analogs, and computational chemistry continue to push the boundaries of possibility, aiding us in perfecting and optimizing venom-based treatments with unprecedented accuracy. As we continue to unlock nature's enigma of destruction and rebirth, it is the duty of the scientific community to invert the esoteric essence of this topic into something easily accessible and understandable to the public. This approach will lead to substantial funding, which will not only spread awareness about the topic, but can make it cheap, easily scalable, and available to a greater population. As far as we know, the possibilities are endless, and it is now up to a new generation of scientists to tread gently into that good night and uncover its secrets.

Works Cited

- Bordon, K. de C. F., Cologna, C. T., Fornari-Baldo, E. C., Pinheiro-Júnior, E. L., Cerni, F. A., Amorim, F. G., Anjolette, F. A. P., Cordeiro, F. A., Wiesel, G. A., Cardoso, I. A., Ferreira, I. G., Oliveira, I. S. de, Boldrini-França, J., Pucca, M. B., Baldo, M. A., & Arantes, E. C. (2020). From Animal Poisons and Venoms to Medicines: Achievements, Challenges and Perspectives in Drug Discovery. *Frontiers in Pharmacology*, 11. <https://doi.org/10.3389/fphar.2020.01132>
- Chen, Y., Pitzemberger, S. M., Garsky, V. M., Lumma, P. K., Sanyal, G., & Baum, J. (1991). Proton NMR assignments and secondary structure of the snake venom protein echistatin. *Biochemistry*, 30(50), 11625–11636. <https://doi.org/10.1021/bi00114a004>
- Halm, D. (2024, June 24). *Neglected Tropical Diseases and Long-Term Captive Animals: Ethical Considerations with Venom Lab Snakes*. *Journal of Applied Animal Ethics*

Research; University of California San Diego.

<https://philarchive.org/archive/HALNTD>

Harvey, A. L. (2014). Toxins and drug discovery. *Toxicon*, 92, 193–200.

<https://doi.org/10.1016/j.toxicon.2014.10.020>

Hawgood, B. J. (1999). Doctor Albert Calmette 1863–1933: founder of antivenomous serotherapy and of antituberculous BCG vaccination. *Toxicon*, 37(9), 1241–1258.

[https://doi.org/10.1016/s0041-0101\(99\)00086-0](https://doi.org/10.1016/s0041-0101(99)00086-0)

Ibraheem, D., Elaissari, A., & Fessi, H. (2014). Administration strategies for proteins and peptides. *International Journal of Pharmaceutics*, 477(1-2), 578–589.

<https://doi.org/10.1016/j.ijpharm.2014.10.059>

Lazarovici, P., Marcinkiewicz, C., & Lelkes, P. I. (2019). From Snake Venom's Disintegrins and C-Type Lectins to Anti-Platelet Drugs. *Toxins*, 11(5).

<https://doi.org/10.3390/toxins11050303>

Li, L., Huang, J., & Lin, Y. (2018). Snake Venoms in Cancer Therapy: Past, Present and Future. *Toxins*, 10(9). <https://doi.org/10.3390/toxins10090346>

Marte, F., & Cassagnol, M. (2022, January 21). *Captopril*. PubMed; StatPearls Publishing.

<https://www.ncbi.nlm.nih.gov/books/NBK535386/>

Oliveira, A. L., Viegas, M. F., da Silva, S. L., Soares, A. M., Ramos, M. J., & Fernandes, P.

A. (2022). The Chemistry of Snake Venom and Its Medicinal Potential. *Nature Reviews Chemistry*, 6(6), 1–19. <https://doi.org/10.1038/s41570-022-00393-7>

Ojeda, P., Ramírez, D., Alzate-Morales, J., Caballero, J., Kaas, Q., & González, W. (2017).

Computational Studies of Snake Venom Toxins. *Toxins*, 10(1), 8.

<https://doi.org/10.3390/toxins10010008>

Rafferty, J. (2019). 9 of the World's Deadliest Snakes | Britannica. In *Encyclopædia*

Britannica. <https://www.britannica.com/list/9-of-the-worlds-deadliest-snakes>

- Santos, L., Oliveira, C., Vasconcelos, B. M., Vilela, D., Melo, L., Ambrósio, L., da Silva, A., Murback, L., Kurissio, J., Cavalcante, J., Cassaro, C. V., Barros, L., Barraviera, B., & Ferreira, R. S. (2020). Good management practices of venomous snakes in captivity to produce biological venom-based medicines: achieving replicability and contributing to pharmaceutical industry. *Journal of Toxicology and Environmental Health, Part B*, 24(1), 30–50. <https://doi.org/10.1080/10937404.2020.1855279>
- Thomas, L. (2021, December 22). *The Medical Uses of Venom*. News-Medical.net. <https://www.news-medical.net/health/The-Medical-Uses-of-Venom.aspx>
- Warrell, D. A., Davidson NMdD, null, Greenwood, B. M., Ormerod, L. D., Pope, H. M., Watkins, B. J., & Prentice, C. R. (1977). Poisoning by bites of the saw-scaled or carpet viper (*Echis carinatus*) in Nigeria. *The Quarterly Journal of Medicine*, 46(181), 33–62. <https://pubmed.ncbi.nlm.nih.gov/866568/>
- World Health Organization. (2024). *The top 10 causes of death*. World Health Organization. <https://www.who.int/news-room/fact-sheets/detail/the-top-10-causes-of-death>
- Ying Fang, Orekhov, G., & F. Lerner, Z. (2022, March 12). *Venom-Inspired Medicine: Ancient Chemicals Offer Novel Solutions*. IEEE Pulse. <https://www.embs.org/pulse/articles/venom-inspired-medicine-ancient-chemicals-offer-novel-solutions/>